

Update to Migraine Therapy: New Injectable Preventative Medications

Andrew Tran, PharmD
PGY-1 Pharmacy Resident
May 23rd, 2019

Learning Objectives

- ▶ Review updated guidelines in regards to treatment and prevention of migraines.
- ▶ Evaluate new injectable migraine preventative therapies and their mechanism of action.
- ▶ Discuss place in therapy and identify patients that are candidates for the new injectables.

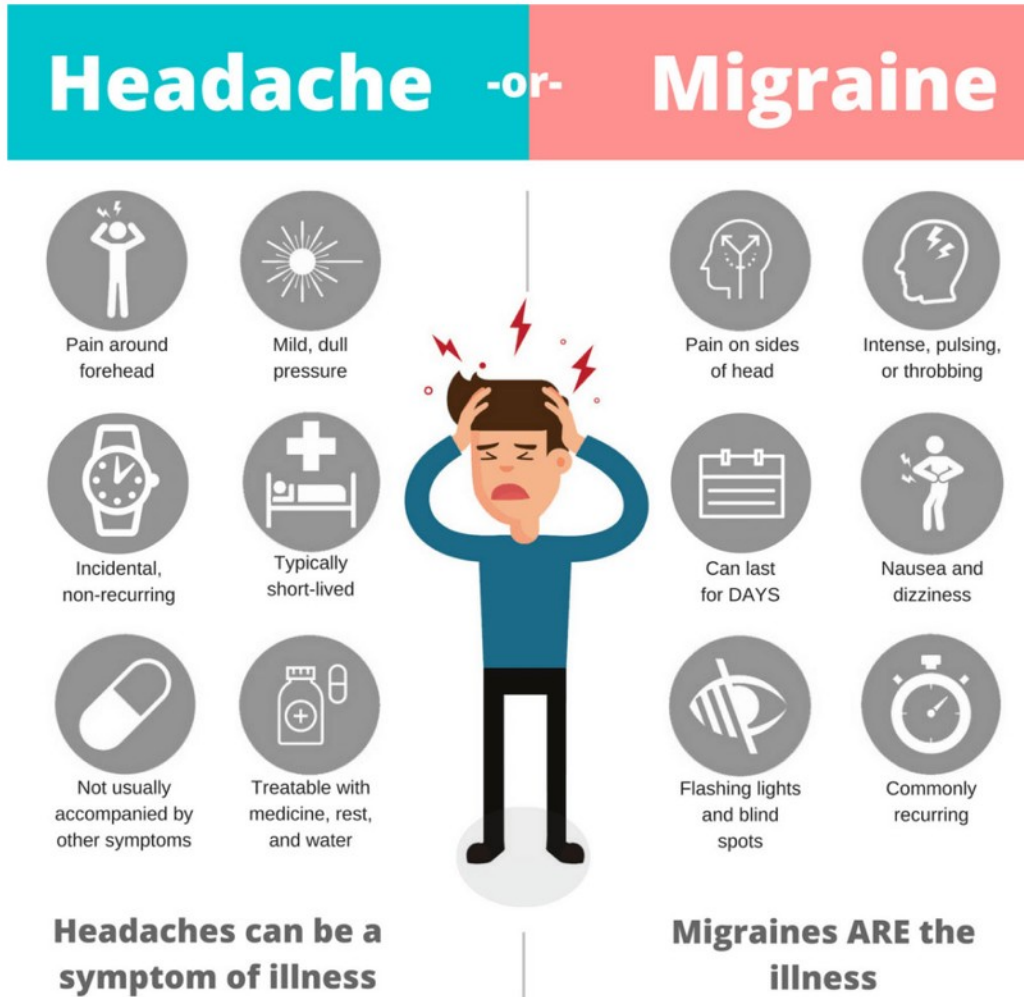
Background

- ▶ Migraines rank as the 3rd most common and 6th most disabling illness.
- ▶ Affects nearly 40 million individuals in the U.S.
- ▶ Females > males (3:1)
- ▶ More than 90% of sufferers are unable to work or function normally during migraines.
- ▶ Estimated healthcare and lost productivity costs ~ \$36 billion annually.

Migraine Characteristics

- ▶ Characterized by:
 - ▶ Chronic severe headaches
 - ▶ Unilateral
 - ▶ Aura/No aura
 - ▶ Pain can last hours or days
 - ▶ Nausea/Vomiting
 - ▶ Sensitivity to light, smells, sounds

Headache or Migraine?

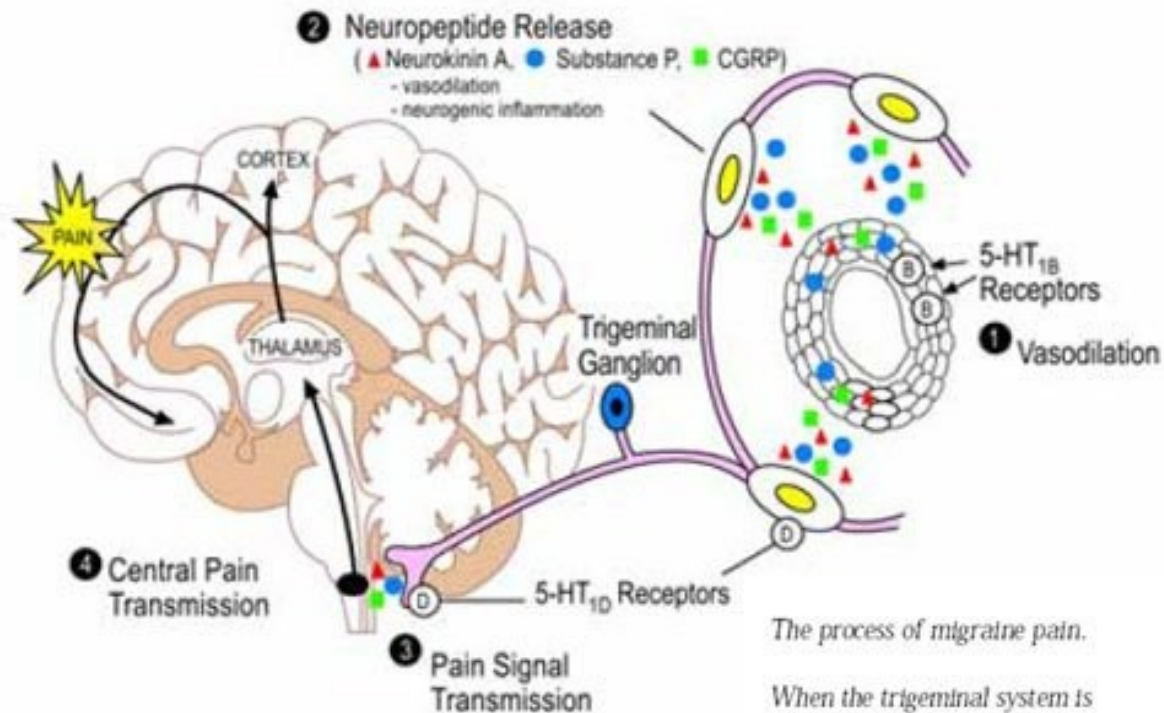


axonoptics
therapeutic eyewear

Risk Factors

- ▶ Family History of Migraine
- ▶ Female gender
- ▶ Stress
- ▶ Menstrual Cycle/Hormones
- ▶ Sleep pattern disruption
- ▶ Diet (skipped meals, alcohol, chocolate cheese, caffeine overuse, MSG, artificial sweeteners, etc.)
- ▶ Medications: estrogens, vasodilators

Migraine Pathogenesis

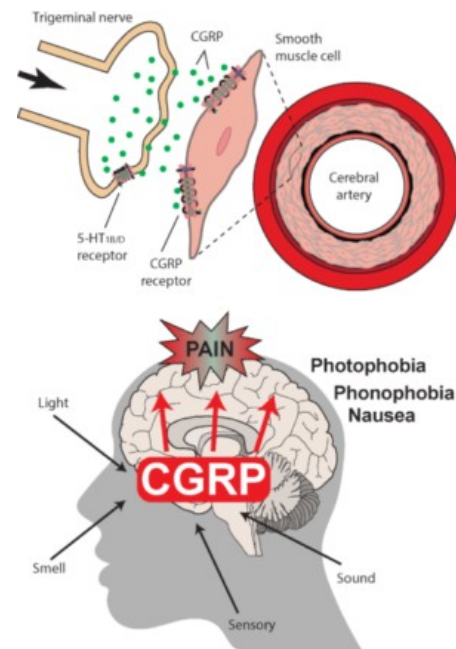


The process of migraine pain.

When the trigeminal system is activated (1), peptides are released (2) prompting an inflammatory reaction. This increases flow of sensory traffic through the brain stem (3), the thalamus and ultimately the cortex (4).

CGRP and Migraines

- ▶ Calcitonin gene-related peptide (CGRP)
 - ▶ Potent vasodilator
 - ▶ Linked in the transmission of pain
 - ▶ Increased levels of CGRP have been reported during migraine attacks



Episodic and Chronic Migraine

Table 1.—ICHD-3 Criteria for Episodic and Chronic Migraine¹

Episodic migraine

- A. At least 5 attacks fulfilling criteria B–D
- B. Headache attacks lasting 4–72 hours (when untreated or unsuccessfully treated)
- C. Headache has at least 2 of the following 4 characteristics:
 - 1. Unilateral location
 - 2. Pulsating quality
 - 3. Moderate or severe pain intensity
 - 4. Aggravation by or causing avoidance of routine physical activity (eg, walking or climbing stairs)
- D. During headache at least 1 of the following:
 - 1. Nausea and/or vomiting
 - 2. Photophobia and phonophobia
- E. Not better accounted for by another diagnosis

Chronic migraine

- A. Migraine-like or tension-type-like headache on ≥ 15 days/month for > 3 months that fulfill criteria B and C
- B. Occurring in a patient who has had at least 5 attacks fulfilling criteria B–D for migraine without aura and/or criteria B and C for migraine with aura
- C. On ≥ 8 days/month for > 3 months, fulfilling any of the following:
 - 1. Criteria C and D migraine without aura
 - 2. Criteria B and C for migraine with aura
 - 3. Believed by the patient to be migraine at onset and relieved by a triptan or ergot derivative
- D. Not better accounted for by another diagnosis

ICHD, International Classification of Headache Disorders.

Acute Treatment Options

- ▶ Triptans
- ▶ Ergot derivatives
- ▶ Dopamine-receptor antagonists
- ▶ Non-opioid analgesics
 - ▶ Tylenol
 - ▶ NSAIDs
- ▶ Opioids
- ▶ Corticosteroids

Indications for Preventive Treatment

- ▶ Attacks significantly interfere with patients' daily routines despite acute treatment
- ▶ Frequent attacks (≥ 4 MHDs)
- ▶ Contraindication to, failure, or overuse of acute treatment defined as:
 - ▶ ≥ 10 days per months for ergot derivatives, triptans, opioids, and combination analgesics
 - ▶ ≥ 15 days per month for nonopioid analgesics, APAP, and NSAIDs
- ▶ Inability to tolerate acute treatments
- ▶ Patient preference

Prophylactic treatments

Table 4.—Treatments With Evidence of Efficacy in Migraine Prevention (Adapted from Silberstein et al¹⁹)

Established efficacy [†]	Probably effective [‡]	Possibly effective [§]
Antiepileptic drugs	Antidepressants	ACE inhibitors: Lisinopril
Divalproex sodium	Amitriptyline	Alpha-agonists
Valproate sodium	Venlafaxine	Clonidine
Topiramate	Beta-blockers	Guanfacine
Beta-blockers	Atenolol	Antiepileptic drugs: Carbamazepine
Metoprolol	Nadolol	Beta-blockers
Propranolol		Nebivolol
Timolol		Pindolol
Triptans: Frovatriptan [¶]		Antihistamines: Cyproheptadine
OnabotulinumtoxinA ³²		Angiotensin receptor blockers: Candesartan

ACE, angiotensin-converting enzyme.

[†]More than 2 Class I trials based on AAN Scheme for Classification of Evidence.³³

[‡]One Class I or 2 Class II studies based on AAN Scheme for Classification of Evidence.³³

[§]One Class II study based on AAN Scheme for Classification of Evidence.³³

^{||}Not for use in women of childbearing potential who are not using an appropriate method of birth control.^{34,35}

[¶]Short-term prevention of menstrual migraine.

^{‡‡}For prevention of chronic migraine.

Indications for anti-CGRP injectables

- ▶ Patient is at least 18 years old
- ▶ Diagnosis of migraine with or without Aura for at least 4 monthly headache days:
 - ▶ Inability to tolerate or inadequate response during 6-week trial of at least TWO other approved preventive therapies
- ▶ Diagnosis of chronic migraine with or without Aura:
 - ▶ Inability to tolerate or inadequate response during 6-week trial of at least TWO other approved preventive therapies

Anti-CGRP Agents

- Erenumab (Aimovig)
- Galcanezumab (Emgality)
- Fremanezumab (Ajovy)

Erenumab (Aimovig®)

- ▶ Approved for: Preventative treatment of migraines in Adults.
- ▶ MOA: binds to the calcitonin gene-related peptide (CGRP) receptor and antagonizes CGRP receptor function.
- ▶ Dose: **70 mg or 140 mg SC monthly**



Aimovig Trials

STRIVE¹ Phase III Study



STRIVE (NCT02456740) is a **6-month** Phase III study evaluating the safety and efficacy of Aimovig 70mg and 140mg in people experiencing between four and 14 migraine days per month.**



121 clinical trial sites in 13 countries



Enrolled **955** patients with an average of **8.3** migraine days per month at baseline

PRIMARY ENDPOINT MET

-3.2 and **-3.7** migraine days per month with Aimovig at 70mg and 140mg doses respectively, **-1.8** migraine days per month for placebo (p<0.001 for both doses vs. placebo)

ARISE² Phase III Study



ARISE (NCT02483585) is a **3-month** Phase III study evaluating the safety and efficacy of Aimovig 70mg in people experiencing between four and 14 migraine days per month.***



76 clinical trial sites in 8 countries



Enrolled **577** patients with an average of **8** migraine days per month at baseline

PRIMARY ENDPOINT MET

-2.9 migraine days per month with Aimovig 70mg, **-1.8** migraine days per month for placebo (p<0.001 vs. placebo)

Chronic Migraine — Phase II Study³



The Pivotal Phase II (NCT02066415) **3-month** study is evaluating the safety and efficacy of Aimovig 70mg and 140mg in people experiencing 15 or more migraine days per month.***



69 clinical trial sites in 10 countries



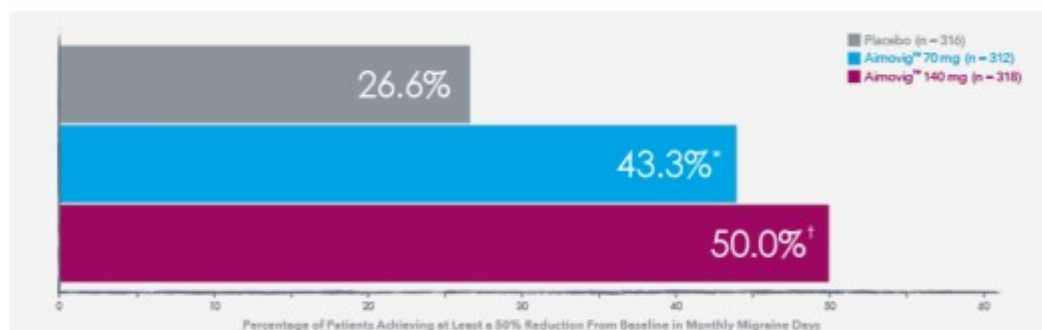
Enrolled **667** people with an average of **18** migraine days per month at baseline

PRIMARY ENDPOINT MET

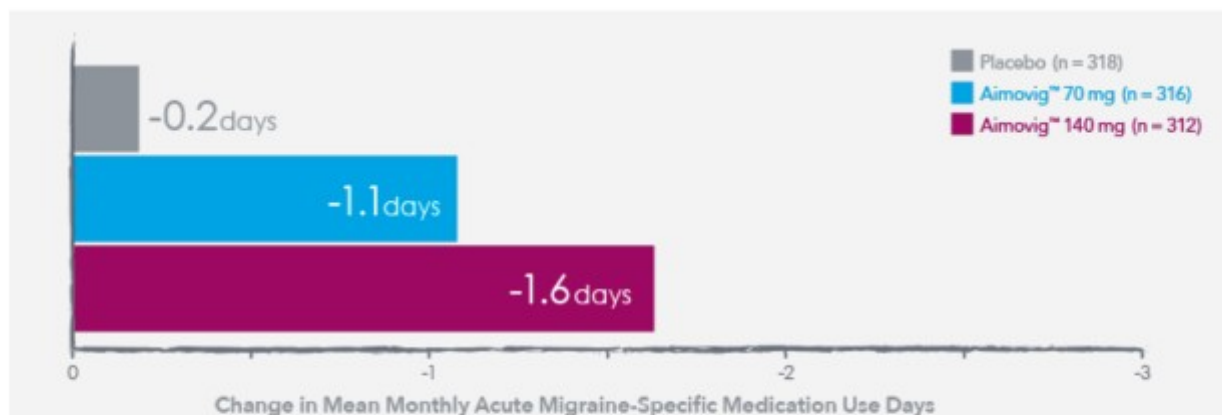
-6.6 migraine days per month with Aimovig at both doses (70mg and 140mg), **-4.2** migraine days per month for placebo (p<0.001 for both doses vs. placebo)

Secondary Outcome Measures

Proportion of patients achieving at least a 50% reduction in mean monthly migraine days



Change from Baseline in Mean Monthly Acute Migraine-specific Medication Days during Months 4-6



Aimovig Adverse Events

Adverse reactions occurring with an incidence of at least 2% for either dose of Aimovig™ and at least 2% greater than placebo through week 12 ¹			
	Placebo (n = 890)	Aimovig™ 70 mg QM (n = 787)	Aimovig™ 140 mg QM (n = 507)
Injection site reactions*	3%	6%	5%
Constipation	1%	1%	3%
Cramps, muscle spasms	< 1%	< 1%	2%

*Injection site reactions include multiple preferred terms, such as injection site pain and injection site erythema.

Galcanezumab (Emgality®)

- ▶ Approved for: Preventative treatment of migraines in Adults.
- ▶ MOA: binds to calcitonin gene-related peptide (CGRP) ligand and blocks its binding to the receptor.
- ▶ Dose: **240 mg SQ loading dose followed by 120 mg SQ monthly**

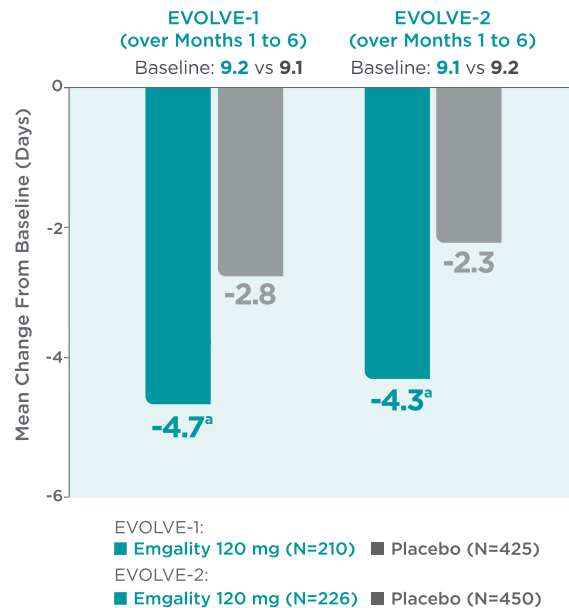


Emgality Trials

EVOLVE-1 (North America) and EVOLVE-2 (Global) were 6-month, randomized, multicenter, double-blind, placebo-controlled studies in patients with episodic migraine

Primary Outcome

Mean Reduction in Monthly MHDs¹



Secondary Outcome Measures



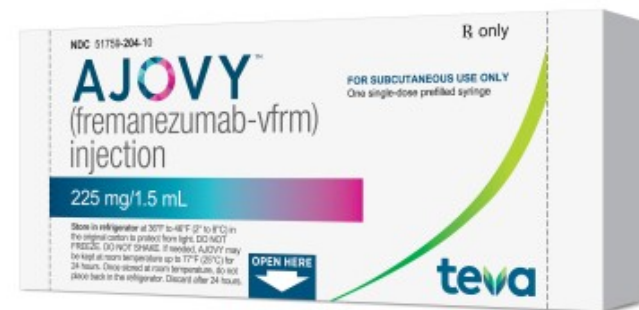
Up to **62%** of patients had a **≥50% reduction** in monthly MHDs from baseline ($p < 0.001$)¹

Up to **39%** of patients achieved a **≥75% reduction** in monthly MHDs from baseline ($p < 0.001$)¹

Up to **16%** of patients had a **100% reduction** in monthly MHDs from baseline ($p < 0.001$)¹

Fremanezumab (Ajoovy®)

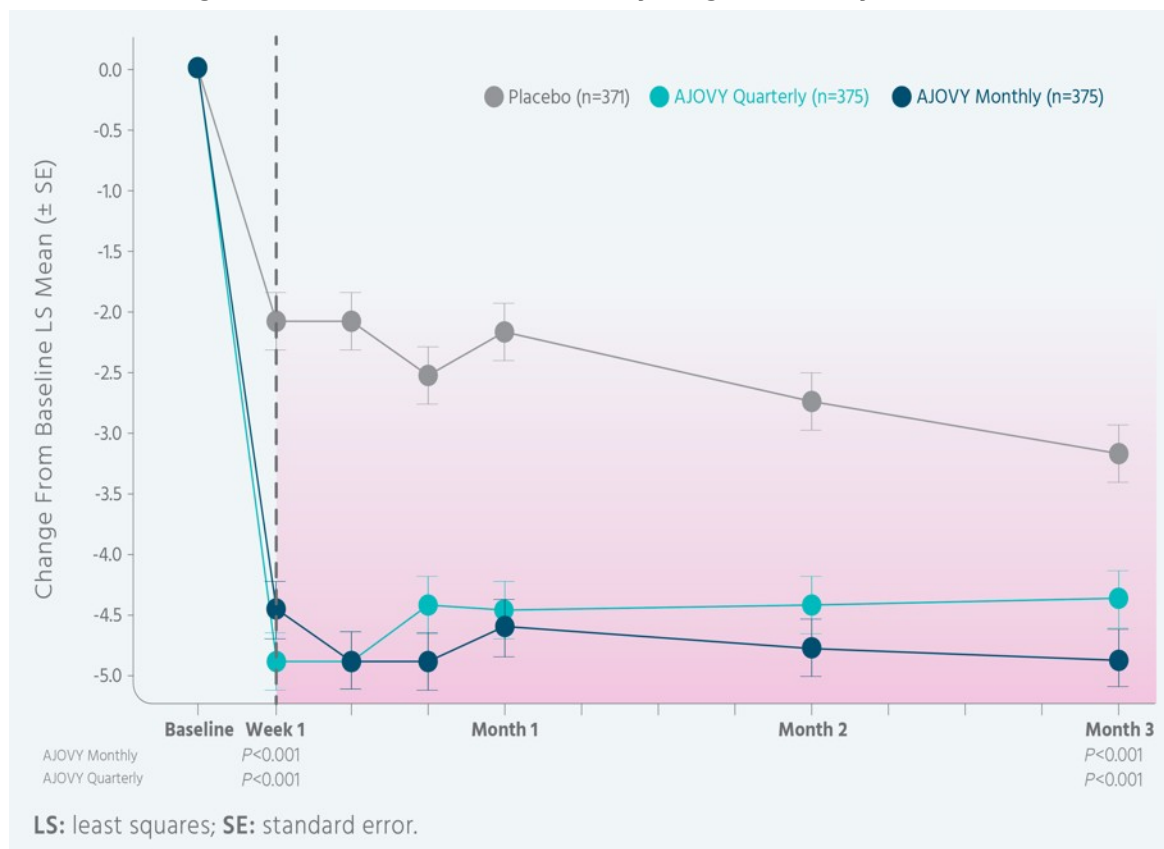
- ▶ Approved for preventative treatment of migraines in adults.
- ▶ MOA: binds to calcitonin gene-related peptide (CGRP) ligand and blocks its binding to the receptor.
- ▶ Dose: **225 mg subQ monthly, or 675 mg subQ every 3 months**



Ajovy Trials

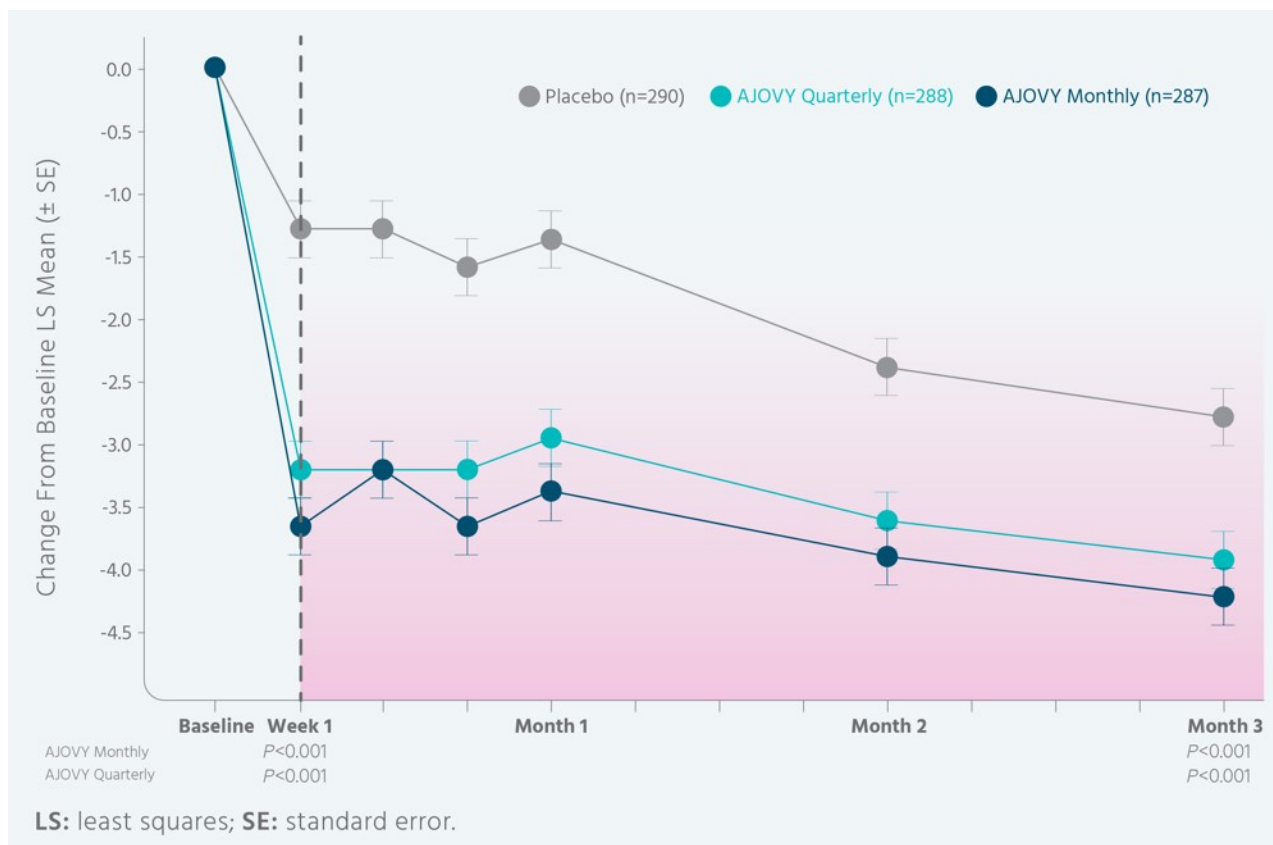
AJOVY was evaluated in two phase 3, 12-week, randomized, double-blind, placebo-controlled, parallel-group trials.

Chronic Migraine: Reduction in Monthly Migraine Days



Ajovy Trials

Episodic Migraine: Reduction in Monthly Migraine Days



New Injectables Overview

	Erenumab (Aimovig®)	Galcanezumab (Emgality®)	Fremanezumab (Ajovy®)
Dosing	Monthly SubQ • 70mg OR • 140 mg	Monthly SubQ • 240 mg loading dose • 120 mg monthly	Monthly or quarterly SubQ • 225 mg monthly • 675 mg Q 3 months
Target	CGRP receptor	CGRP peptide or ligand	CGRP peptide or ligand
Studied For	Episodic, and Chronic Migraine	Episodic, Chronic, episodic cluster headache, and chronic cluster headache	Episodic, Chronic, episodic cluster headache, and chronic cluster headache
Approved on	05/17/18	09/26/18	09/14/18
AWP (per pen)	\$690.00	\$690.00	\$460.00



Key Points

- ▶ New anti-CGRP injectables are viable options after patient has failed other preventative therapies.
- ▶ Choice of which injectable based on patient preference:
 - ▶ Loading dose → Emgality
 - ▶ Quarterly dosing → Ajovy
- ▶ Barriers:
 - ▶ High price
 - ▶ Lack of long-term studies regarding efficacy and safety

Question

- ▶ Which of the following is a valid indication for initiating anti-CGRP mAb?
 - A. Acute treatment of episodic migraines
 - B. Non-responsive to acute treatment
 - C. Inability to tolerate or inadequate response to at least 2 oral preventative therapies
 - D. Migraine in children

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Question

- ▶ Which of the following are barriers to initiating anti-CGRP mAB? (Select 2 of the following)
 - A. Cost
 - B. Numerous adverse effects in Studies
 - C. Frequent Dosing
 - D. Lack of long-term efficacy and safety data

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 - A. Cost
 - B. Numerous adverse effects in Studies**
 - C. Frequent Dosing
 - D. Lack of long-term efficacy and safety data**

References

1. Migraine Facts. Migraine Research Foundation website. <http://migraineresearchfoundation.org/about-migraine/migraine-facts/>. Accessed April 15th, 2019.
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6. Product Information: EMGALITY(TM) subcutaneous injection, galcanezumab-gnlm subcutaneous injection. Eli Lilly and Company (per FDA), Indianapolis, IN, 2018
7. Product Information: AJOVY(TM) subcutaneous injection, fremanezumab-vfrm subcutaneous injection. Teva Pharmaceuticals USA Inc (per FDA), North Wales, PA, 2018.
8. James A. Charles and Alan M. Rapoport, The American Headache Society's Position Statement on Integrating New Migraine Treatments into Clinical Practice – Comments, *Headache: The Journal of Head and Face Pain*, **59**, 4, (629-629), (2019).



Questions?



Andrew Tran, PharmD
Contact: atran@pchc.com